

On the Invariant Theory of Forms in n Variables

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§ 7. Expansion of the General Forms in Terms of Normal Forms.

In what follows (§ 8) we shall derive a principal property of the folding process (41), namely its composability from fundamental foldings. For this we need to transfer to the n -ary domain an important theorem given by Mertens¹ in the quaternary domain. This theorem says that any finite system of forms can be replaced by an equivalent system of normal forms. By a “normal form” we mean—in a generalization of the binary and ternary definition²—a form all of whose invariant formations linear in the coefficients vanish identically, with the exception of the ground form itself. Thus, by Theorem VI (§ 6), a normal form φ satisfies the system of differential equations

$$\left(q_{n-\sigma-\lambda} \frac{\partial}{\partial p_\sigma} \middle| \frac{\partial}{\partial q_{n-\tau}} p_{\tau-\lambda} \right) \varphi = 0, \quad (\sigma \geq \tau, \lambda = 1, 2, \dots, \min(\tau, n - \sigma)), \quad (46)$$

where the rows $q_{n-\sigma-\lambda}, p_{\tau-\lambda}$ are to be regarded as arbitrary quantities. In the quaternary domain—taking (39) into account for $\sigma = \tau = 2$ —these differential equations agree completely with the ten differential equations given by Mertens without introducing the arbitrary rows p, q (loc. cit., formula 28). Their knowledge, together with Theorem VI, allows us to transfer his proof almost word for word to n variables. The only addition is the verification, obtained from the decomposition identity (30), that the constructed forms are normal forms; in the quaternary domain this verification follows without further transformation. It therefore suffices to give a short account of the proof, and, for simplicity, only in the case needed below: a single ground form f and its form row (cf. the end of § 6). The proof remains the same for invariant formations of arbitrary degree in the coefficients of arbitrarily many ground forms.

The basic idea of the proof is to use the transformed coefficients to replace a form with $n - 1$ rows p_ρ by forms with n rows ξ cogredient with the x , namely the rows of the transformation quantities. The sought normal forms are obtained directly from these forms by invariant differentiation processes. The expansion of the general forms in terms of normal forms is mediated by a series expansion for forms with ρ ($\rho \leq n$) cogredient rows ξ ;³ invariant differentiation processes lead back to the forms in the original rows p_ρ .

Let $\xi^{(1)}, \dots, \xi^{(n)}$ be the rows of the transformation quantities, so that

$$\begin{aligned} x_i &= \xi_i^{(1)} x'_1 + \xi_i^{(2)} x'_2 + \dots + \xi_i^{(n)} x'_n, \\ p_{i_1 \dots i_\rho} &= \sum_j (\xi_{i_1}^{(j_1)} \dots \xi_{i_\rho}^{(j_\rho)}) p'_{j_1 \dots j_\rho}. \end{aligned} \quad (47)$$

Denote the transformed coefficients of the forms $f, \varphi, \dots$ by $(f), (\varphi), \dots$, and let

$$f = (S^1 | p_1)^{m_1} (S^2 | p_2)^{m_2} \dots (S^{n-1} | p_{n-1})^{m_{n-1}}.$$

Then

$$\begin{aligned} (f) &= (S^1 | \xi^{(1)})^{m_{11}} \dots (S^1 | \xi^{(n)})^{m_{1n}} (S^2 | \xi^{(1)} \xi^{(2)})^{m_{21}} \dots (S^{n-1} | \xi^{(2)} \dots \xi^{(n)})^{m_{n-1,n}} \\ &\quad \cdot (s^{(1)} | \xi^{(1)})^{k_1} (s^{(2)} | \xi^{(2)})^{k_2} \dots (s^{(n)} | \xi^{(n)})^{k_n}, \end{aligned} \quad (48)$$

where $m_{11} + \dots + m_{1n} = m_1$, and so on. From each coefficient (f) for which

$$k_1 \geq k_2 \geq k_3 \geq \dots \geq k_n, \quad (49)$$

the differentiation process that exactly exhausts the rows ξ ,

$$\left(\frac{\partial}{\partial \xi^{(1)}} \middle| p_1 \right)^{k_1 - k_2} \left(\frac{\partial}{\partial \xi^{(1)}} \frac{\partial}{\partial \xi^{(2)}} \middle| p_2 \right)^{k_2 - k_3} \dots \left(\frac{\partial}{\partial \xi^{(1)}} \dots \frac{\partial}{\partial \xi^{(n-1)}} \middle| p_{n-1} \right)^{k_{n-1} - k_n} \left(\frac{\partial}{\partial \xi^{(1)}} \dots \frac{\partial}{\partial \xi^{(n)}} \right)^{k_n} \quad (50)$$

¹F. Mertens, *Über invariante Gebilde quaternärer Formen* (see the Introduction), cited as “Mertens.”

²Study, loc. cit., p. 54.

³The current R823 source gives $\rho < n$ at line 4121. The original print has $\rho \leq n$; formula (54) and the later explicit treatment of $\rho = n$ confirm that the endpoint is included. The inclusive printed reading is therefore restored and disclosed here.

gives the form

$$\varphi = (s^{(1)} | p_1)^{k_1-k_2} (s^{(1)} s^{(2)} | p_2)^{k_2-k_3} \dots (s^{(1)} \dots s^{(n-1)} | p_{n-1})^{k_{n-1}-k_n} (s^{(1)} s^{(2)} \dots s^{(n)})^{k_n}. \quad (51)$$

The forms φ are the sought normal forms. Indeed, they have the following properties that define normal forms.

1. They are invariant formations of the ground form f that are linear in the coefficients. This follows because they arise from f through the invariant differentiation processes $\left(\frac{\partial}{\partial p_\rho} \Big| \xi^{(i_1)} \xi^{(i_2)} \dots \xi^{(i_\rho)}\right)$ and (50). Thus, by the end of § 6, they can be expressed linearly through polars of the finite form row of f . These polars are obtained by regarding the variables p_ρ entering φ as coefficients of a form decomposed into linear forms with the variable rows p_ρ :

$$H = (q_{n-1} | p_{n-1})^{k_1-k_2} (q_{n-2} | p_{n-2})^{k_2-k_3} \dots (q_1 | p_1)^{k_{n-1}-k_n}. \quad (52)$$

The simultaneous invariants of the form rows f and H give all the polars under consideration, since the latter must have the same dimension in each row p_ρ as φ itself.

2. They satisfy the differential equations (46). If the differential process (45) is applied to φ , then

$$\varphi' = (q_{n-\sigma-\lambda} s^{(1)} \dots s^{(\sigma)} | [s^{(1)} \dots s^{(\tau)}]_{n-\tau} p_{\tau-\lambda}), \quad (\sigma \geq \tau).$$

From the decomposition identity (30), if the rows $q_\sigma = s^{(1)} \dots s^{(\sigma)}$ and $p_{n-\tau} = [s^{(1)} \dots s^{(\tau)}]_{n-\tau}$ formed from the rows s are identified with the q_ρ, p_τ occurring there, it follows that

$$\varphi' = \sum_{\beta=\sigma-\tau+\lambda}^{n-\tau,\sigma} \sum_{i_\beta} \varepsilon(\dot{q}_{\sigma-\beta}^{(i_1 \dots i_\beta)} q_{n-\tau+\lambda} | p_{\sigma+\lambda} \dot{p}_{n-\tau-\beta}^{(i_1 \dots i_\beta)}).$$

The original print marks both complementary symbols in this double sum with dots. The current R823 source drops both dots. The printed readings $\dot{q}$ and $\dot{p}$ are restored here.⁴ In this case,

$$\dot{p}_{n-\tau-\beta}^{(i_1 \dots i_\beta)} \sim s^{(1)} \dots s^{(\tau)} s^{(i_1)} \dots s^{(i_\beta)},$$

where $i_1, \dots, i_\beta$ denote any β distinct numbers selected from the numbers 1 through σ . The original print again has $\dot{p}$ in the preceding display, whereas R823 substitutes p' . The printed dotted symbol is restored.⁵ Since $\beta > \sigma - \tau$, some of the numbers $i_1, \dots, i_\beta$ must lie between 1 and τ ; consequently $\dot{p}_{n-\tau-\beta}^{(i_1 \dots i_\beta)}$ vanishes identically,⁶ and hence every individual matrix product, and therefore also φ' , vanishes.

For the equivalence of the system of normal forms with the form row f , it remains only to show that all forms of the form row can be expanded in polars of the normal forms in the sense specified under 1. Let Π be an aggregate of polar operations

$$\left(\frac{\partial}{\partial \xi^{(i)}} \Big| \xi^{(k)}\right), \quad (i \neq k; i = 1, 2, \dots, \rho; k = 1, 2, \dots, \rho), \quad (53)$$

and let Π^σ be an aggregate of the special operations (53) for which $i = \sigma$ and $k < \sigma$. For a form F of the ρ rows $\xi^{(1)}, \dots, \xi^{(\rho)}$ that has degree k_ρ in the row $\xi^{(\rho)}$, the following expansion holds⁷:

$$F = \sum_{\alpha=0}^{k_\rho} \Pi F_\alpha, \quad (\rho \leq n), \quad (54)$$

where F_α has degree α in the last row $\xi^{(\rho)}$ and arises from F through the invariant differential operations

$$\left(\frac{\partial}{\partial \xi^{(1)}} \dots \frac{\partial}{\partial \xi^{(\rho)}} \Big| \xi^{(1)} \dots \xi^{(\rho)}\right)^\alpha \quad (55)$$

and Π^ρ . If, in constructing F_α , process (55) is replaced by

$$\left(\frac{\partial}{\partial \xi^{(1)}} \dots \frac{\partial}{\partial \xi^{(\rho)}} \Big| p_\rho\right)^\alpha, \quad (56)$$

⁴The two restored readings are separately tracked in the accompanying source-defect and formula ledgers.

⁵Specialist interpretation of this dotted complementary symbol remains external-review debt.

⁶The original print retains the dot in this prose occurrence; R823 drops it. The printed reading is restored here.

⁷F. Mertens, *Über eine Formel der Determinantentheorie*. Sitzungsber. d. Ak. d. Wiss., Wien, math.-naturw. Kl., Bd. 91, 2. Abt. (1885), formula 20).

then a form $F_\alpha(p_\rho)$ with only $\rho - 1$ rows $\xi^{(1)}, \dots, \xi^{(\rho-1)}$ is obtained, to which (54) can again be applied. Continuing the procedure leads to forms $G(p_\rho, p_{\rho-1}, \dots, p_1)$. To recover the expansion of F in (54) from these forms, one replaces each row p_ρ by $\xi^{(1)}, \dots, \xi^{(\rho)}$ and applies the polar process Π in (53) to the resulting forms; this gives (cf. (48)) a sum of transformed coefficients (G). For $\rho = n$, process (55) remains in the first step. If c denotes numerical constants and, for brevity, we put

$$(\xi^{(1)}\xi^{(2)}\dots\xi^{(n)}) = \Delta; \quad \left(\frac{\partial}{\partial\xi^{(1)}} \frac{\partial}{\partial\xi^{(2)}} \dots \frac{\partial}{\partial\xi^{(n)}} \right) = \nabla, \quad (57)$$

then, in the case $\rho = n$,

$$F = \sum c\Delta^\alpha(G), \quad (58)$$

whereas for $\rho < n$ only $\alpha = 0$, and hence $\sum c(G)$, occurs on the right.

Applying (58) to a transformed coefficient (f) in (48), and using the commutativity of the polar processes Π^σ with all operations (56) for which $\rho \geq \sigma$, as well as the fact that polars of transformed coefficients again give transformed coefficients, we obtain

$$G = \left(\frac{\partial}{\partial\xi^{(1)}} \middle| p_1 \right)^{\beta_1} \left(\frac{\partial}{\partial\xi^{(1)}} \frac{\partial}{\partial\xi^{(2)}} \middle| p_2 \right)^{\beta_2} \dots \left(\frac{\partial}{\partial\xi^{(1)}} \dots \frac{\partial}{\partial\xi^{(n-1)}} \middle| p_{n-1} \right)^{\beta_{n-1}} \nabla^{\beta_n} \Pi(f) = \sum c\varphi,$$

and from (58) it follows that

$$(f) = \sum c\Delta^\alpha(\varphi) \quad (\text{Mertens, formula 23}). \quad (59)$$

To pass from (59) to the expansion of f itself, we again regard the variables p_ρ as coefficients of a form H in (52), with the degrees $m_1, \dots, m_{n-1}$ corresponding to f , and put $m = m_1 + \dots + m_{n-1}$. Then, by the definition of invariants,

$$f \cdot \Delta^m = \sum (f)(H) = \sum c\Delta^\alpha(\varphi)(H),$$

and application of the process ∇^m that exactly exhausts the rows ξ gives

$$f = \sum c\Phi, \quad (60)$$

where the forms Φ are derived from φ and H by invariant differentiation processes with respect to the variables, and hence are simultaneous invariants of the form rows φ and H .⁸ Since φ is a normal form, however, its form row reduces to the form φ itself; the form row H contains every possible invariant combination of variables. Thus the simultaneous invariants of φ and H are the polars of φ in the sense specified under 1; that is, (60) gives the expansion of f in polars of the normal forms. It remains to verify the same expansion for an arbitrary form θ of the form row f . Let Γ denote the normal forms derived from (θ) by (50), so that

$$(\theta) = \sum c\Delta^\beta(\Gamma). \quad (61)$$

The forms θ are characterized by the relation valid for every transformed coefficient,

$$(\theta)\Delta^\sigma = \sum c(f) \quad (\text{Mertens, formula 9}).$$

But every form F of the n rows ξ that contains the factor Δ^σ also contains the second $\nabla^\sigma \Pi F$ (Mertens, formula 20); hence

$$(\theta) = \sum c\nabla^\sigma(f), \quad \text{and therefore} \quad \Gamma = \sum c\varphi,$$

and from (61) we obtain the relation corresponding to (59),

$$(\theta) = \sum c\Delta^\beta(\varphi),$$

which implies for θ the expansion (60) in polars of the normal forms φ . To summarize:

Theorem VII. Every form row f can be replaced by an equivalent system of normal forms.

⁸Instead of the direct conclusion, Mertens, § 7, introduces a series of further differential processes whose essential purpose is to construct the form row H ; this is accomplished by our Theorem VI.